

QIAN MA

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🎓 EDUCATION

Rensselaer Polytechnic Institute

Jan. 2024 – Jan. 2028 (expected)

Ph.D. Student in Computer Science, advised by [Prof. Yao Ma](#)

City University of Hong Kong

2022 – 2024

M.Sc. in Multimedia Information Technology, Distinction

University of Electronic Science and Technology of China

2018 – 2022

B.Eng. in Software Engineering - Systems and Technology

⚙️ RESEARCH INTERESTS AND EXPERTISE

- Current research interests and expertise: Multimodal Learning and Reasoning, Vision-Language Models (VLMs), Multimodal RAG, Knowledge-Based Visual Question Answering (KB-VQA), and reliable foundation-model evaluation.
- Other expertise: Graph Learning, Foundation Models for Graphs, Graph Self-Supervised Learning, spatio-temporal forecasting, time-series modeling, and urban computing.

👥 RESEARCH EXPERIENCE

DAMI-Lab, Rensselaer Polytechnic Institute

Jan. 2024 – Present

Ph.D. Researcher Advisor: [Prof. Yao Ma](#)

- Lead first-author projects on multimodal RAG and KB-VQA, studying how VLM/MLLM systems ground visual queries to external knowledge [1][2].
- First-authored ACL26 paper [2], showing that KB-VQA systems benefit from separating entity identification from evidence selection instead of treating retrieval, grounding, and ranking as one entangled step.
- First-authored ECCV26 paper [1], showing that common evaluation results can be distorted by unsupported answers, underspecified questions, and visually simplified single-entity settings.
- First-authored work towards foundation model for graph on graph SSL scaling/evaluation [3][4] and LLM-guided graph structure learning [5].
- My research trajectory moved from graph foundation models toward multimodal learning, RAG, and KB-VQA because both directions center on how foundation models ground non-textual structure, visual entities, and external knowledge.

AML Lab, City University of Hong Kong

Oct. 2022 – Jun. 2023

Research Assistant Mentor: [Zijian Zhang](#); Advisors: [Prof. Xiangyu Zhao](#), [Prof. Haoliang Li](#)

- First-authored HIRST (CIKM 2023) [6], showing that traffic forecasting benefits from modeling sensor dependencies through regional and global hierarchies rather than only local sensor-level patterns.
- Developed the method, experiments, and paper with advisors; the project was accepted by CIKM 2023 and received an A+ for the graduate dissertation project.
- Collaborated on *PromptST* [7] (CIKM 2023), covering prompt learning and multi-attribute time-series prediction for smart-city applications.

University of Electronic Science and Technology of China

Dec. 2019 – Jun. 2020

Undergraduate Research Assistant Supervisor: Prof. Su Sheng

- Processed large-scale urban driving records for grid-based urban computing research and implemented MapReduce-style parallel preprocessing to utilize all CPU cores.

INDUSTRY RESEARCH AND ENGINEERING EXPERIENCE

AT&T Chief Data Office, Bedminster, NJ

May 2025 – Aug. 2025

Graduate Research Intern Mentors: [Qiong Wu](#), [Zhengyi Zhou](#); Manager: [Divesh Srivastava](#)

- Conducted research on multimodal RAG for knowledge-grounded visual question answering, focusing on how systems should identify visual entities before retrieving and reasoning over external evidence.
- Initiated research directions that led to first-authored ECCV 2026 and ACL 2026 papers on KB-VQA benchmark reliability and workflow design [1][2].
- Built reproducible datasets, prompts, evaluation code, and retrieval baselines to support internal research and public release.

SAP Labs China, Xi'an

Apr. 2021 – May 2022

Software Engineering Intern, SAP Data Intelligence Manager: [Jingtao Li](#)

- Developed and maintained two CI/CD automation microservices for a cloud infrastructure service automation platform.
- Worked with SaaS and cloud-native teams on infrastructure workflows, and presented new projects/features in internal global sharing sessions for SAP T&I HANA Database and Analytics.

SELECTED PUBLICATIONS

- [1] **Qian Ma**, S M Rayeed, Charles V. Stewart, Qiong Wu, Yao Ma. *Identifying and Resolving Pitfalls of Knowledge-Based VQA Benchmarks: Auditing, Repairing, and Augmenting*. ECCV 2026, to appear.
- [2] **Qian Ma**, Qiong Wu, Zhengyi Zhou, Yao Ma. *Ground Then Rank: Revisiting Knowledge-Based VQA with Training-Free Entity Identification*. ACL 2026 Findings, to appear.
- [3] **Qian Ma**, Haitao Mao, Jingzhe Liu, Zhehua Zhang, Chunlin Feng, Yu Song, Yihan Shao, Yao Ma. *Do Neural Scaling Laws Exist on Graph Self-Supervised Learning?* LoG 2024.
- [4] **Qian Ma**, Hongliang Chi, Hengrui Zhang, Kay Liu, Zhiwei Zhang, Lu Cheng, Suhang Wang, Philip S. Yu, Yao Ma. *Overcoming Pitfalls in Graph Contrastive Learning Evaluation: Toward Comprehensive Benchmarks*. ACM SIGKDD Explorations, 2025.
- [5] **Qian Ma**, Haitao Mao, Zhehua Zhang, Qiong Wu, Zhengyi Zhou, Yao Ma. *Cross-Domain GraphWalker: Harnessing LLMs for Graph Structure Learning*. Under review.
- [6] **Qian Ma**, Zijian Zhang, Xiangyu Zhao, Haoliang Li, Hongwei Zhao, Yiqi Wang, Zitao Liu, Wanyu Wang. *Rethinking Sensors Modeling: Hierarchical Information Enhanced Traffic Forecasting*. CIKM 2023.
- [7] Zijian Zhang, Xiangyu Zhao, Qidong Liu, Chunxu Zhang, **Qian Ma**, et al. *PromptST: Prompt-Enhanced Spatio-Temporal Multi-Attribute Prediction*. CIKM 2023.

TECHNICAL SKILLS

- Programming: Python, Java, JavaScript, Go, C; research code in PyTorch and PyTorch Lightning.
- Multimodal/RAG: VLM/MLLM evaluation, Qwen-VL, LLaVA, Llama, EVA-CLIP, CLIP, FAISS, BGE reranking, prompt-based evaluation, KB-VQA benchmarks.
- Graph/Time-Series: GNNs, graph self-supervised learning, graph contrastive learning, graph foundation models, spatio-temporal forecasting.
- Systems and Tools: Linux, Git, Docker, Kubernetes, Slurm, Singularity, Jenkins, $\LaTeX$.

HONORS AND AWARDS

Outstanding Student Scholarship

2018 – 2019

Outstanding Student Scholarship

2020 – 2021

Outstanding Student Scholarship

2021 – 2022

ADDITIONAL INFORMATION

- Languages: English - fluent; Mandarin - native.
- English proficiency: IELTS 7.0 (Listening 7.0, Reading 8.5, Writing 6.0, Speaking 5.5); CET-6 551.